

Analysis | Published: 21 November 2016

Tumor-suppressor genes that escape from X-inactivation contribute to cancer sex bias

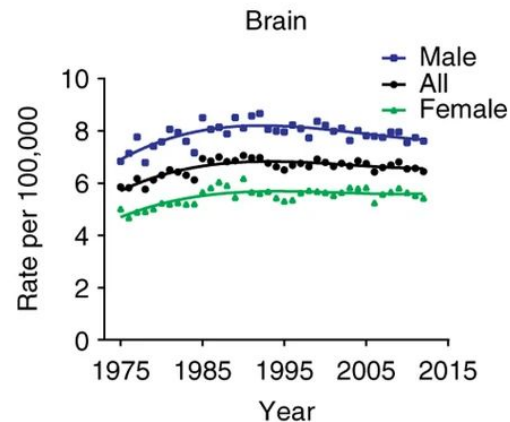
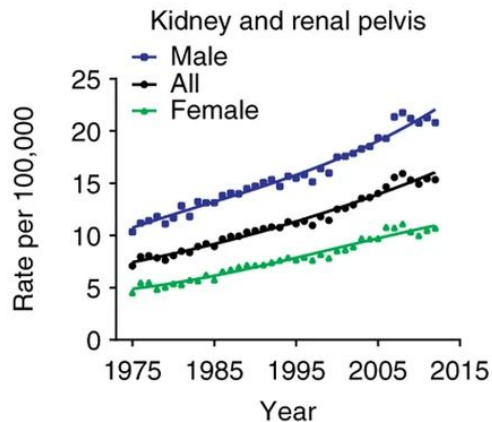
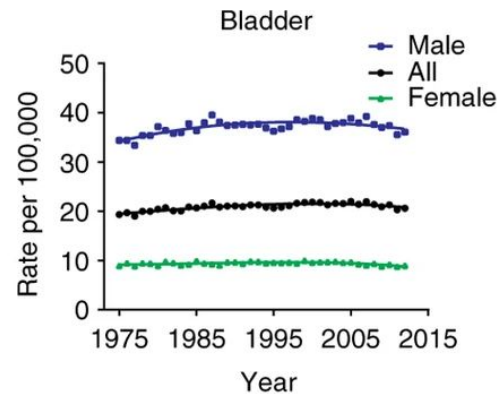
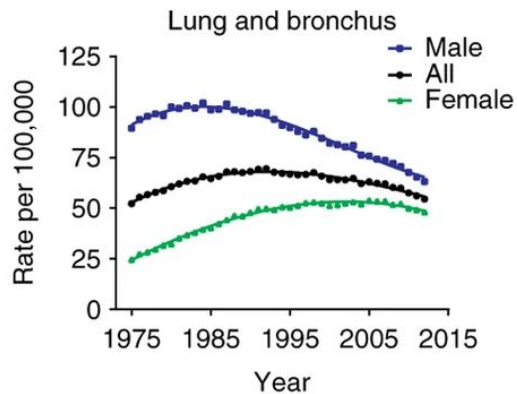
[Andrew Dunford](#), [David M Weinstock](#), [Virginia Savova](#), [Steven E Schumacher](#), [John P Cleary](#), [Akinori Yoda](#), [Timothy J Sullivan](#), [Julian M Hess](#), [Alexander A Gimelbrant](#), [Rameen Beroukhim](#), [Michael S Lawrence](#), [Gad Getz](#) & [Andrew A Lane](#) 

Presenter: Yilu Wang
June 3, 2026

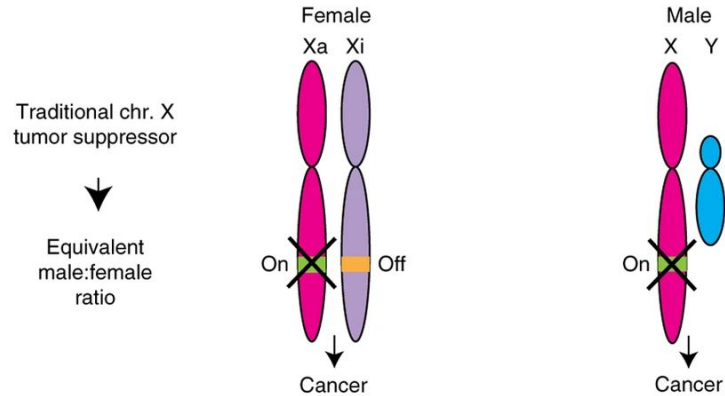


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Male excess across many cancer types remains largely unexplained

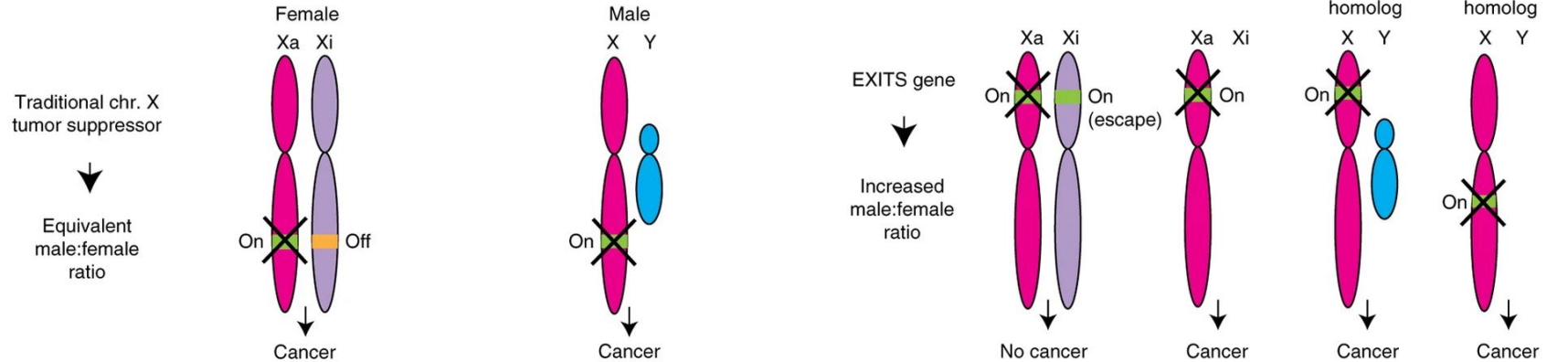


SEER data of
annual incident rates



Clues:

- KDM6A loss-of-function mutations are male-biased
- Female cancers show biallelic loss (Knudson two-hit hypothesis)



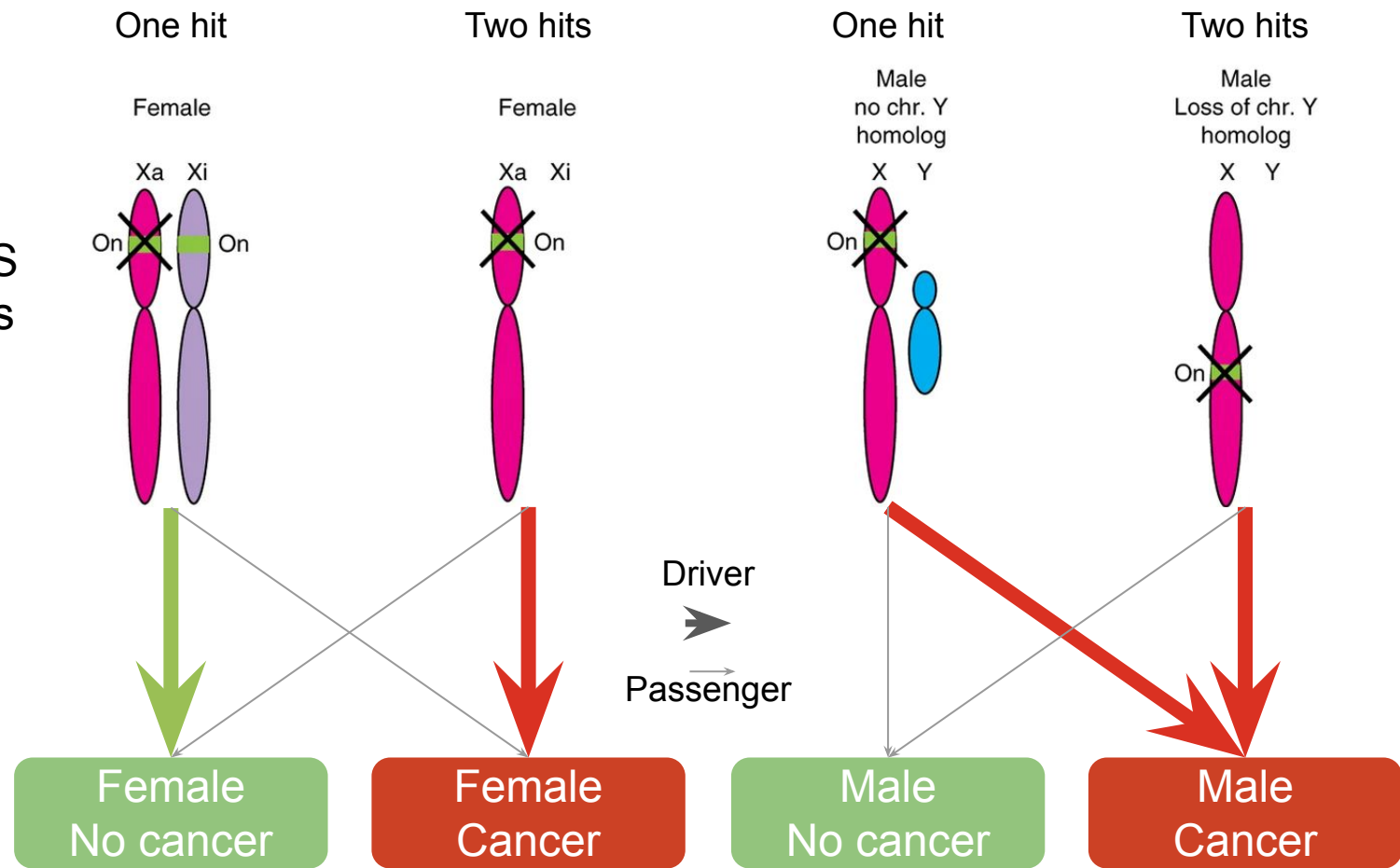
Clues:

- KDM6A loss-of-function mutations are male-biased
- Female cancers show biallelic loss (Knudson two-hit hypothesis)

Hypothesis:

Some X-chr Tumor Suppressor genes Escape X-Inactivation, so females have two active alleles — EXITS genes

EXITS
genes



EXITS
genes

One hit

Female



Two hits

Female



One hit

Male
no chr. Y
homolog



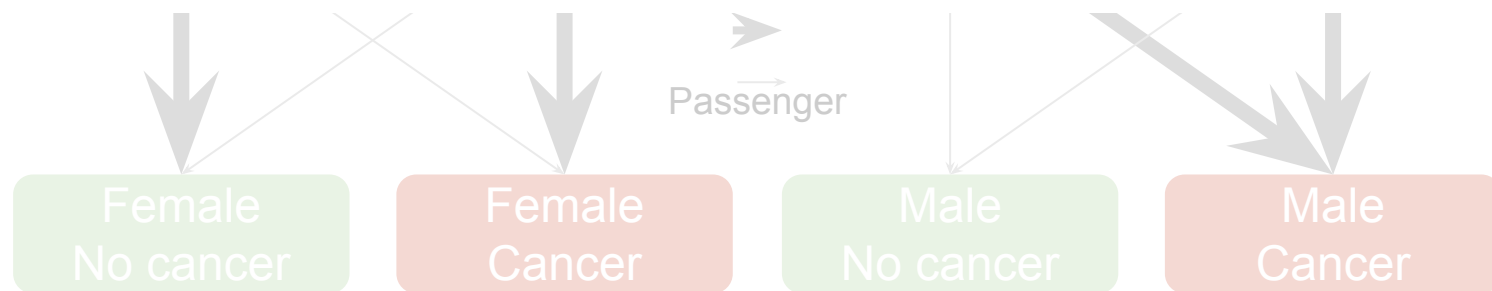
Two hits

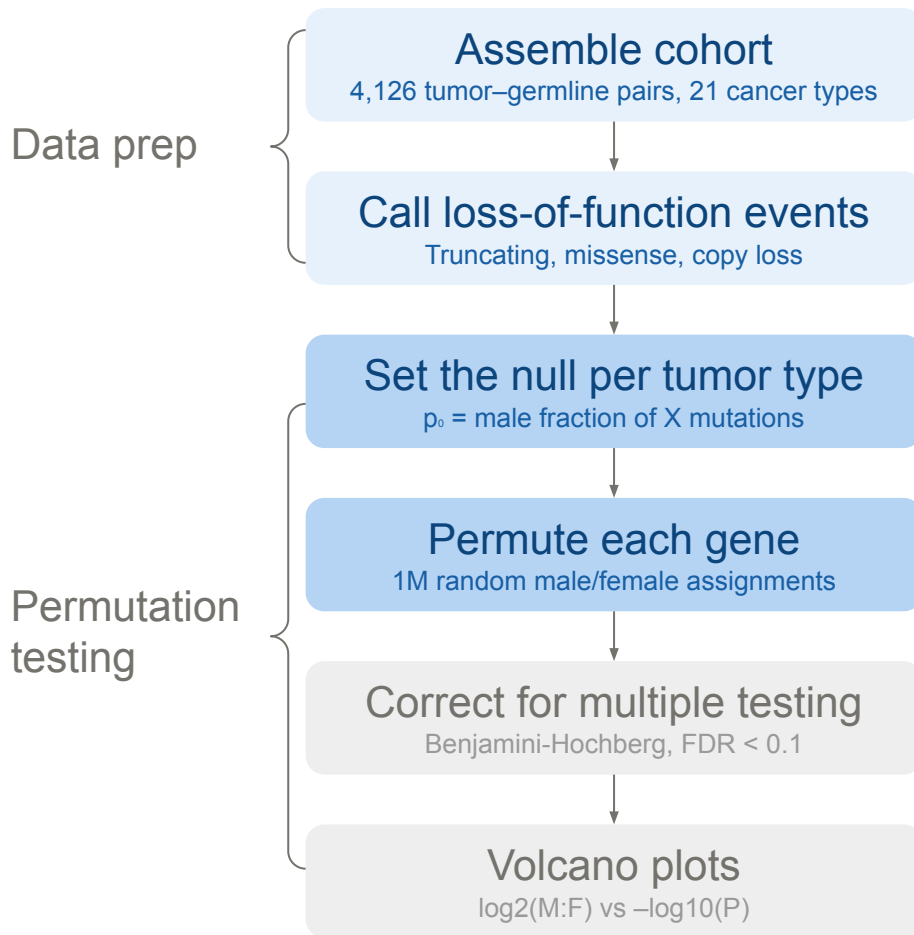
Male
Loss of chr. Y
homolog



Prediction:

Loss-of-function mutations in EXITS genes should be more frequent in male cancers than female after normalization





What fraction of X-chr mutations
should fall in males
just by chance

Data prep

Assemble cohort

4,126 tumor-germline pairs, 21 cancer types

Call loss-of-function events

Truncating, missense, copy loss

Set the null per tumor type

p_0 = male fraction of X mutations

Permute each gene

1M random male/female assignments

Correct for multiple testing

Benjamini-Hochberg, FDR < 0.1

Volcano plots

$\log_2(\text{M:F})$ vs $-\log_{10}(P)$

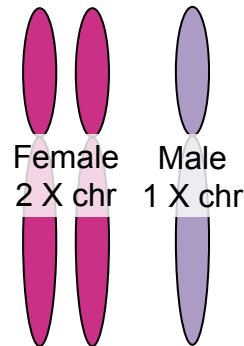
Permutation
testing

Male:Female incidence

98:72

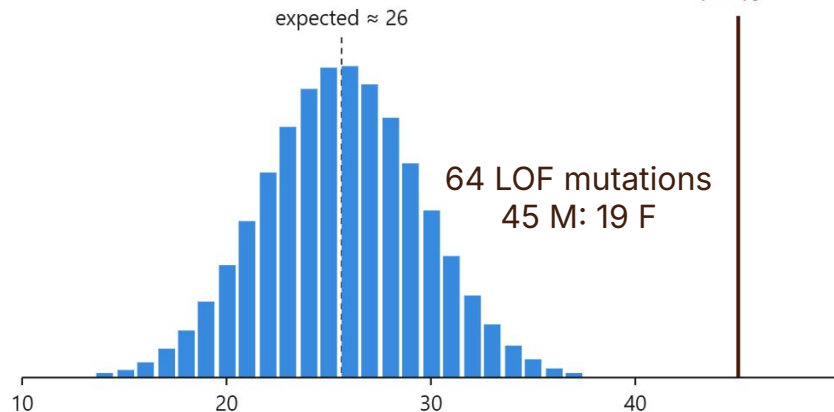
Male: 98 patients × 1 X chr
Female: 72 patients × 2 X chr

Male fraction of
X mutations under null
 $p_0 = 98 / (98 + 72 \times 2) = 0.4$



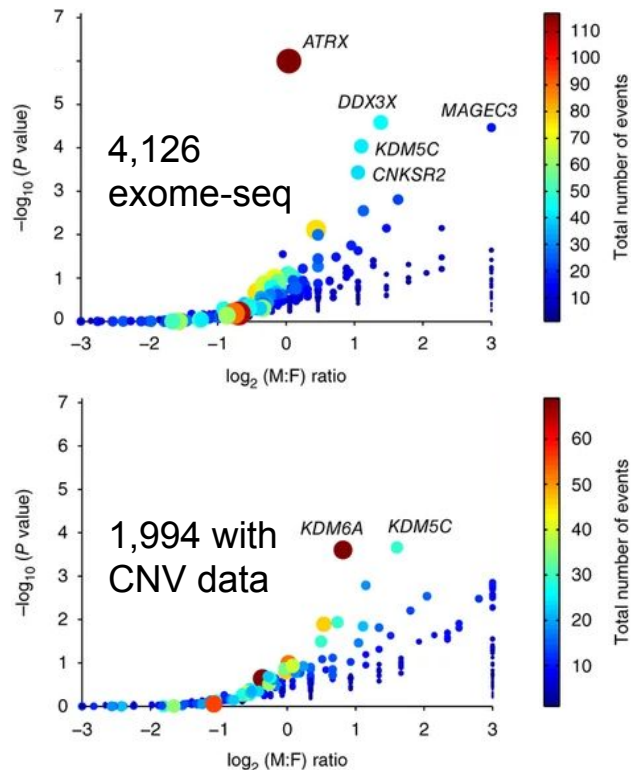
ATRX in low-grade glioma:

observed = 45
 $P \approx 10^{-6}$

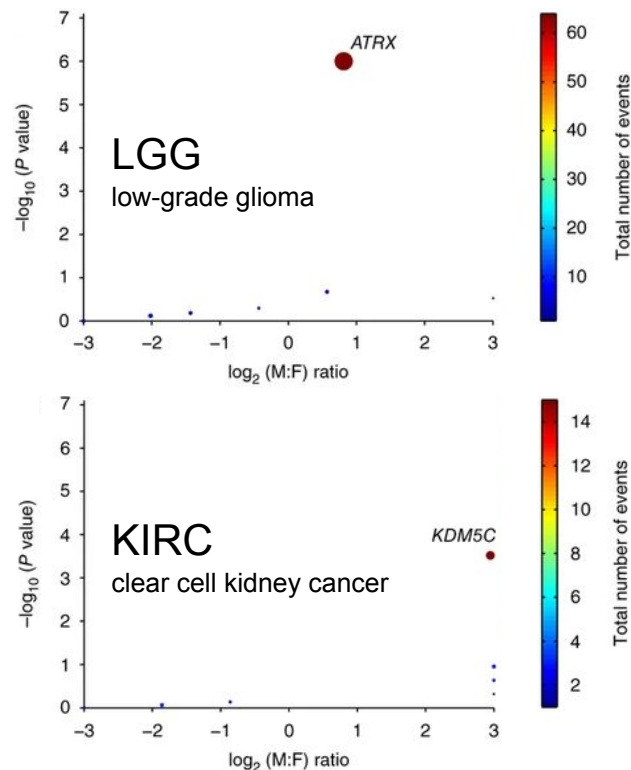


Number of male ATRX events (out of 64 total)

Pan-cancer permutation testing



Disease-specific permutation testing



Six of 783 non-PAR X-chr genes show male-biased loss

ATRX in LGG and KDM5C in KIRC show male-biased loss specifically

Recently recognized escape genes

CNKSR2

MAGEC3

ATRX

Canonical escape genes

DDX3X

KDM5C

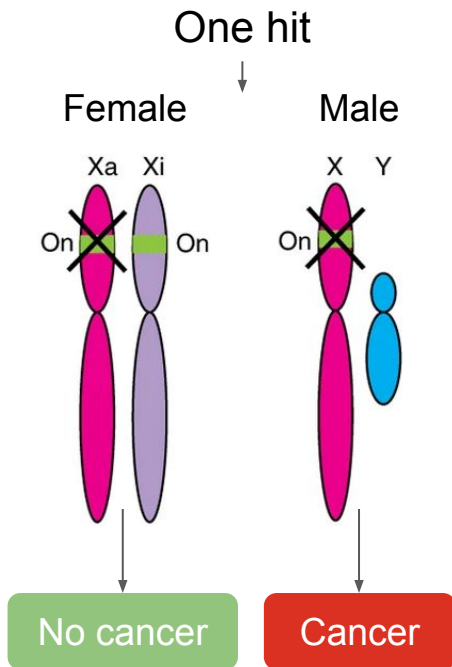
KDM6A

Also with Y homolog

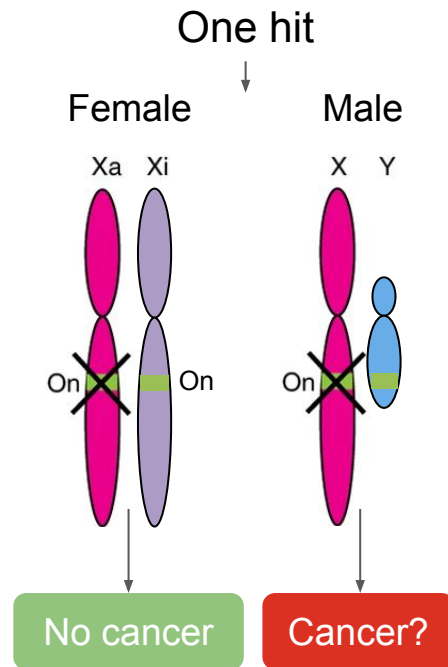
Hypothesis:

ATRX is a context-dependent
escape gene, particularly in the
brain (*supported by GTEx data*)

No Y homolog:
ATRX, CNKSR2, MAGEC3

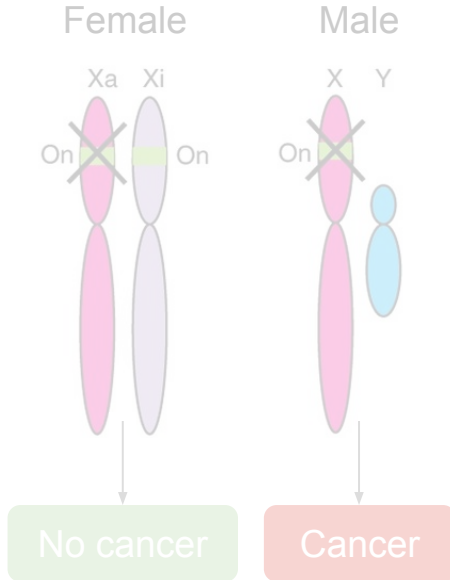


With Y homolog:
DDX3X, KDM5C, KDM6A



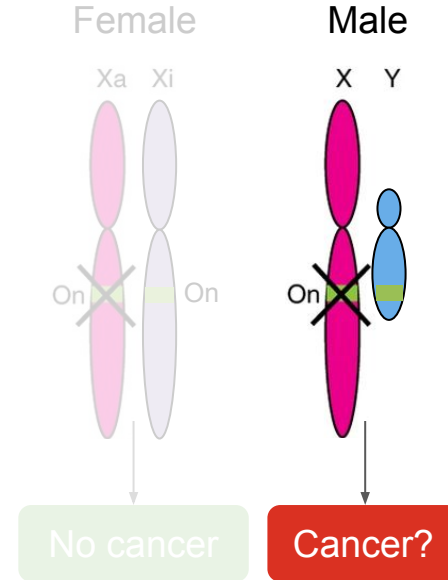
No Y homolog:
ATRX, CNKSR2, MAGEC3

One hit



With Y homolog:
DDX3X, KDM5C, KDM6A

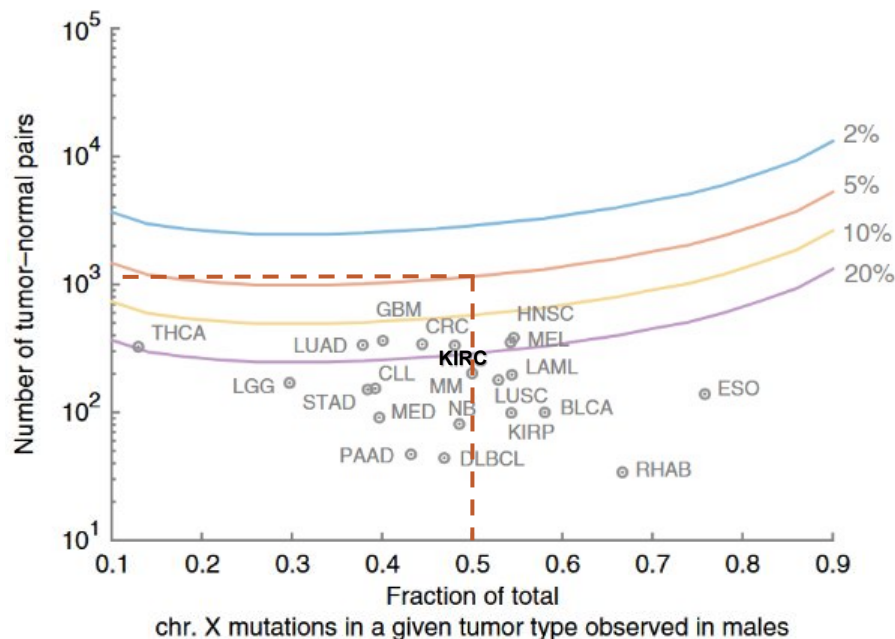
One hit



Hypotheses:

- Y loss pre-exists associated with age and tobacco
- Y counterparts do not have equal tumor-suppressor activity

Most current cohorts are underpowered for low-prevalence EXITS genes

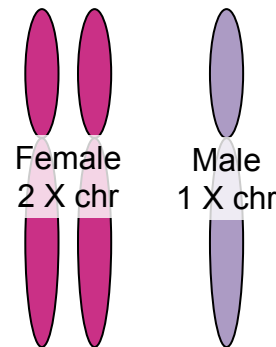


Male:Female incidence

2:1

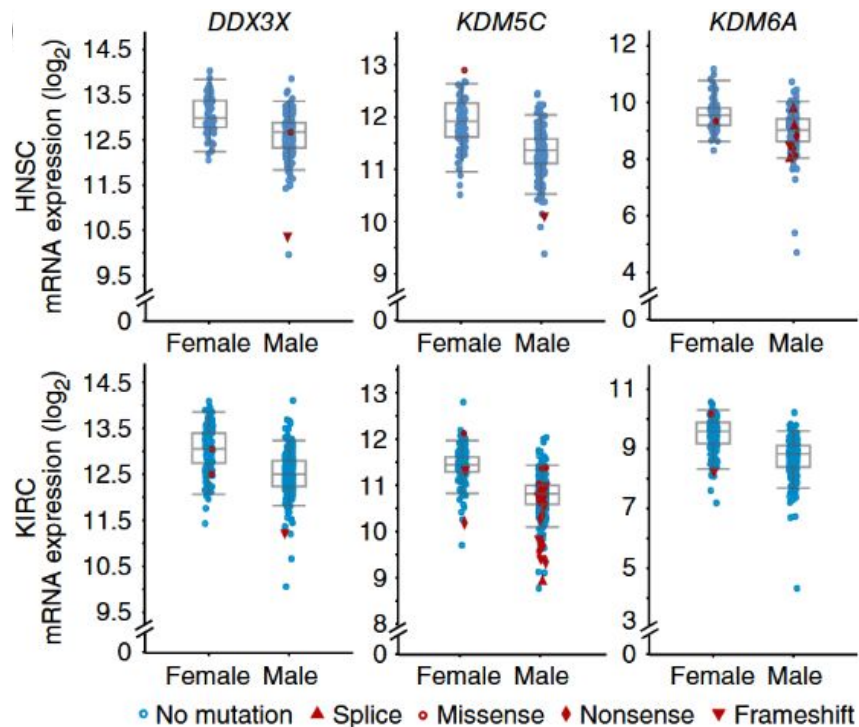
Male: 2 patients × 1 X chr
Female: 1 patient × 2 X chr

Male fraction of X mutations under null
 $p_0 = 2 / (2 + 2 \times 1) = 0.5$



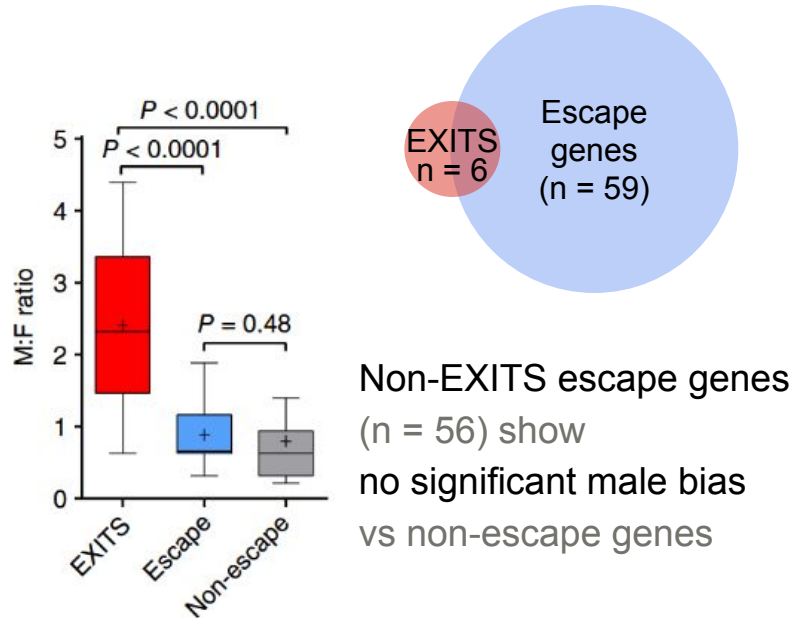
To detect a 4× male bias for a gene mutated in 5% of KIRC cases at 80% power:
> 1,000 tumors needed (current cohort: 334)

Male bias in EXITS genes extends beyond coding mutations

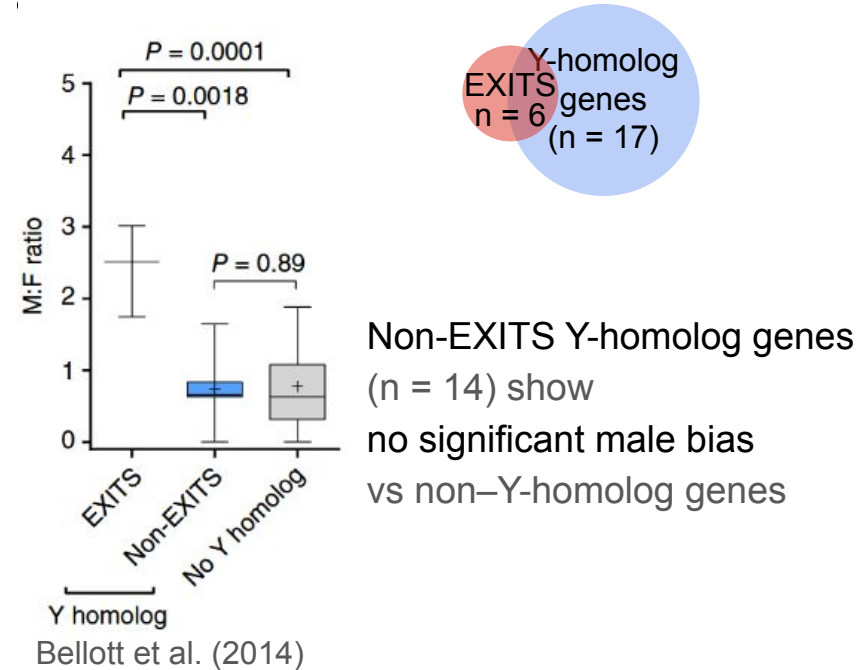


- Females express EXITS genes higher than males, consistent with X-escape
- In tumors without coding mutations, the lowest-expression cases are predominantly male
 - DDX3X: 22M vs 2F ($P = 0.012$)
 - KDM5C: 16M vs 2F ($P = 0.071$)
 - KDM6A: 23M vs 2F ($P = 0.0077$)

Male bias is largely confined to the six EXITS genes



Carrel & Willard (2005)

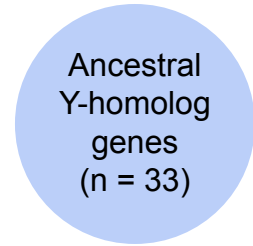


Bellott et al. (2014)

Two additional EXITS genes are identified with targeted screen



RBM10
known as a TSG

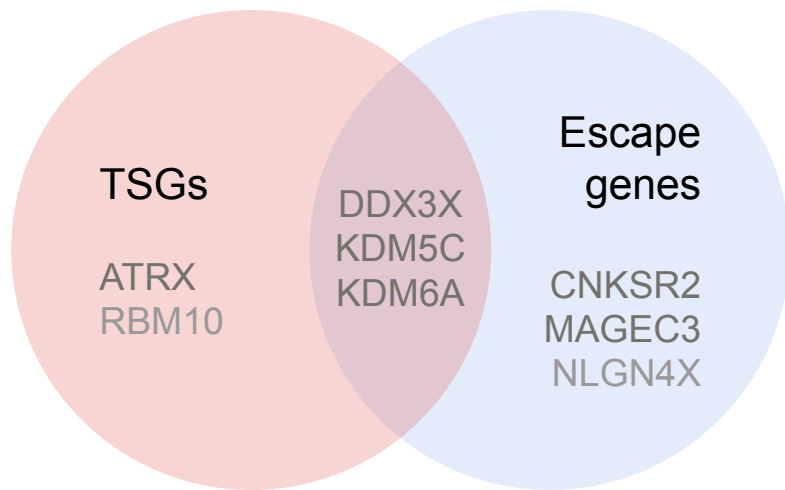


NLGN4X

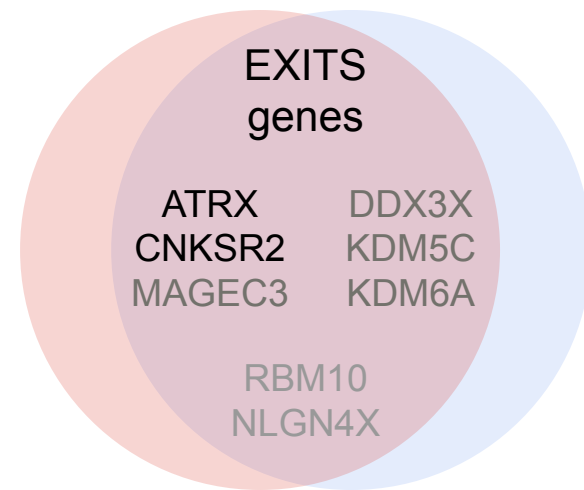
Hypothesis:

EXITS genes which were not recognized (CNKSR2, MAGEC3, and NLGN4X) are candidate tumor suppressors

Before:
scattered prior knowledge



After:
unified as EXITS



Escape X-inactivation and act as tumor suppressors → male-biased loss

Strengths

- Unbiased genome-wide screen

All 783 non-PAR X-chr genes tested with no prior filter

- A network of hypotheses, predictions, and evidence

Male LOF bias, male Y loss, tissue-specific escape (ATR_X in brain), candidate tumor suppressors (CNKSR2), expression evidence, etc.

- Proper statistical normalization

Per-tumor-type p_0 controls for cohort sex ratio and X-dosage

- *Reference-list dependence for targeted screen*

The escape and Y-homolog gene sets are from two prior studies

Limitations

- Underpowered for rare EXITS genes

Most cohorts can't detect genes mutated in < 5% of cases

- Non-coding mechanisms not proven

Non-coding mutations/epigenetic silencing not validated by WGS/WGBS

- Cis/trans ambiguity

Short-read exome-seq can't confirm the two LOF mutations are biallelic

- *Mutation-rate assumption*

May not hold for exposure-driven cancers, thus distorting p_0

Builds on

This paper

Extended by

Knudson two-hit hypothesis

TSGs need both copies lost (1971)

Escape-gene catalog

Carrel & Willard (2005)

Y-homolog dosage genes

Bellott et al. (2014)

Isolated cases

e.g. KDM6A in T-ALL (2009)

**EXITS: a unifying
mechanism for the
male excess**

Dunford et al. (2017)

**Whole-genome/
non-coding drivers**

Li et al. (2018); PCAWG Consortium (2020)

EXITS now canonical

Reviews (2020+)

Clinical prediction markers

e.g., inactive-X loss flags worse or better
prognosis in different cancers (2020+)

Thank you!

Presenter: Yilu Wang
June 3, 2026



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